UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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|---|---|
| COURSE | DATABASE DESIGN AND DEVELOPMENT |
| COURSE CODE | ICT450 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 6 printed pages.*

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## QUESTION 1 (20 marks)

a) A database management system (DBMS) is a software tool that supports various activities and functions for data management. Briefly explain TWO (2) functions of DBMS that guarantee integrity and consistency of data.

(4 marks)

b) Identify FOUR (4) examples of database applications usage.

(4 marks)

c) Table 1 shows an example of data from Gerai 17's database system. The owner of Gerai 17, Cik Siti, is having difficulty getting the total number of menu and average price of her menu. Identify THREE (3) problems exist in Table 1. Justify your answer.

(6 marks)

Table 1: MENU Table

| Menu Code | Menu Name | Price (RM) |
|---|---|---|
| M1 | Nasi Bujang | 3.50 |
| M2 | Nasi Ayam | |
| | Bihun Goreng | 3.00 |
| M4 | Laksa | 2.S0 |

d) Database system is designed based on two design philosophies which are centralized and decentralized. Explain THREE (3) differences between these two designs.

(6 marks)

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**QUESTION 2** **(20 marks)**

a) The evolution of data models has been driven by the increasing complexity of data storage, management, and analysis needs over time. Describe TWO (2) disadvantages of hierarchical model.

(4 marks)

b) Business rules are precise statements, derived from a detailed description of the organization's operations. Identify FOUR (4) sources of business rules.

(4 marks)

c) Backup the database is one of the crucial aspects in Database Life Cycle (DBLC). Describe THREE (3) types of backup.

(6 marks)

d) Briefly explain THREE (3) advantages of mobile database in marketing area.

(6 marks)

## QUESTION 3 (20 marks)

Company IT Creative Solution makes an initiative to record its employees' skills. Table 3 shows a sample record for some of its employees. The company will request up to 2 important skills from each employee. However, this record is still in poor design. Some employees have not passed a certification test in any skill, while others have been certified in several skills. Some certified skills are shared by several employees, while other skills have no employees that hold those certifications.

Table 3 : Employee Skills Records of IT Creative Solution in 2025

| Item | Staff ID | Staff Name | Position | Department ID & Name | Skill ID & Skill Name | Completion Date |
|---|---|---|---|---|---|---|
| 1. | 304 | Intan Syafinaz | Analyst | D1 - Data Analytic | AD - Advanced Database | 1/1/2025 |
| 2. | 304 | Intan Syafinaz | Analyst | D1 - Data Analytic | BM - Basic DB Manipulation | 3/1/2025 |
| 3. | 301 | Robert James | UI/UX Designer | D2 - Software development | AD - Advanced Database | 1/1/2025 |
| 4. | 302 | Mohd Fikri Mohd Ali | Developer | D2 - Software development | - | - |
| 5. | 307 | Zinnirah | Developer | D2 - Software development | AD - Advanced Database | 1/1/2025 |
| 6. | 307 | Zinnirah | Developer | D2 - Software development | FF - Frontend Frameworks | Not complete yet |
| 7. | 305 | Linda Clark | System Admin | D3 - Network and Systems Administration | BM - Basic DB Manipulation | 3/1/2025 |

a) State the normal form of Table 3. Explain your answer with reasons.

(3 marks)

b) Identify the primary key(s) and partial dependencies and/or transitive dependencies.

(6 marks)

c) Normalize Table 1 from first normal form (1NF) until third normal form (3NF). Show all the steps involved.

(11 marks)

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## QUESTION 4 (20 marks)

Table 2 shows the relational schema for a mechanic center that tracks all service jobs.

Table 2: Relational schema for a mechanic center

| |
|---|
| CENTER (Center_ID, Center_Name, Center_City) |
| MECHANIC (Mech_ID, Mech_Name, Mech_Specialty, Mech_Salary, Center_ID*) |
| CUSTOMER (Cust_ID, Cust_Name, Cust_CarModel) |
| SERVICE (Service_ID, Service_Date, Mech_ID*, Cust_ID*) |

Write the SQL statements based on the following requirements:

a) Create the table MECHANIC with a primary key and a foreign key.

(4 marks)

b) List the mechanic name and salary for mechanics specialized in 'ENGINE' and earning more than RM 5 000. Sort the result by salary in descending order.

(4 marks)

c) Display the total number of different customers who received service at centers located in 'Melaka'.

(6 marks)

d) By using a subquery, display the center name, center city, and mechanic name for the mechanic with the highest salary.

(6 marks)

## QUESTION 5 (20 marks)

Collector's Hub is a world renowned toy/memorabilia collectibles shop that was founded in 1980. Collectors from all over the world would buy collectibles from this shop since it specializes in providing rare collectibles. Collectibles such as movie memorabilia, statues, model kits, action figures and others are highly sought after by the collectors if it is rare and have a high appreciation value. The company has been managing its customers' order and sales by manual bookkeeping but, in order to keep up with the technology advancement of today, they have decided to upgrade from their manual system to an online sales database system via a website. After a careful and thoughtful consideration, they have chosen you to develop the online sales database system. Below are the requirements and business rules that the development of the database system must be based on:

- The system needs to keep track of the information on collectors, the collectibles that the company is offering and the staff information.
- The collectors' information such as the name, address, telephone number and their email need to be recorded into the system.
- The collectibles' name, description, price and type are also the information that needs to be tracked by the system.
- The system also needs to record the staffs' name, phone, email and their manager's identification.
- A collector can buy many collectibles and a collectible can be bought by many collectors. Each sale transaction must also record the date, time, quantity sold and the total sale.
- A staff manages at least one sales transaction and a transaction can only be managed by one staff only.
- A manager that is appointed among the staff must monitor at least one staff. A staff can only be monitored by one manager only.

In terms of the collectibles, there are five types of collectibles that needs to be tracked by the system. The collectors can buy the collectibles whether it is memorabilia, statues, model kits, action figures and bricks. Each of the collectibles must have one type only and each of these types have attributes data that needs to be recorded. In terms of memorabilia, the name of the movie is required. As for the rest, type of build is required for statues, scale is for model kits, wave year is for action figures while number of pieces is required for bricks.

Based on the information above, create a complete Entity Relationship Diagram (ERD) that shows all entities, attributes, relationship and connectivities.

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